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Maritime Circular Methanol Cycle, Enabled by Onboard Carbon Capture and Circular Methanol Production on Floaters

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Abstract

This paper examines a sustainable fuel cycle that combines the production of E-fuels with a significant reduction of emissions in the maritime sector by deploying Onboard Carbon Capture on ships with floating production units for circular methanol.

The paper focuses on an integrated cycle in which CO_2 is captured and liquefied onboard shipping vessels, significantly reducing their greenhouse gas emissions. This technology, which has already been successfully piloted and is ready for scale-up and commercial deployment, can be applied to both new and existing ships. Upon arrival in port, the captured CO_2 can be supplied to a methanol production plant that manufactures circular methanol from the CO_2, using the circular CO_2 as a feedstock.

The key design feature of this cycle is that the manufacturing plant, in this case, is designed and built as an integrated Floating Methanol Production and Storage Unit (FMPS). This enables the clean FMPS technology to be available in any port in the world, either to distribute synthetic methanol to maritime shipping or through other synthetic fuel value chains. The floating facility has a number of advantages:

- Relatively rapid construction, deployment and autonomous on-site start-up.
- Beyond established ports, the FMPS could be located at new sites where Renewable Power is potentially available in abundance and produced at a relatively low cost. In these cases, the production of green hydrogen (either on the floating unit or a shore-based facility), coupled with the ship-based captured CO_2 as feedstock, produces a truly carbon-neutral fuel.
- To cover the energy transition period, the FMPS could also be deployed near existing industrial sites where Blue or Grey Hydrogen is produced. In these cases, onshore-produced hydrogen, together with ship based carbon captured CO_2, is used as floater feedstock.
- Relocatable, allowing the entire plant to be moved easily if deployment is required elsewhere.
- Lower cost, improved safety and quality through fabrication in a controlled shipyard environment.

The paper outlines the ship-based Onboard Carbon Capture (OCCS) technology whilst detailing the integrated floater technology, particularly the configuration and deployment options. Case studies, together with supporting economic analysis using the levelized cost of FMPS-produced synthetic fuel production, are presented.

Introduction

Global shipping currently accounts for around 1,000 MTPA of CO_2 emissions, equivalent to around 3% of global annual emissions. This is a significant part of the total global emissions, at a level similar to aviation.

The International Maritime Organization (IMO) is responsible for driving a reduction in these emission levels. In 2018, the IMO adopted an initial strategy on the reduction of GHG emissions from ships, setting out a vision to reduce GHG emissions from international shipping and phase them out as soon as possible.

In July 2023, the IMO adopted the 2023 IMO Strategy on Reduction of GHG Emissions from Ships in accordance with the agreed program of follow-up actions. The 2023 IMO GHG Strategy (IMO, 2023) envisages a reduction in the carbon intensity of international shipping of at least 40% by 2030. It also includes a level of ambition relating to the uptake of zero or near-zero GHG emission technologies, fuels and/or energy sources, which are to account for at least 5% of the energy used by international shipping by 2030.

Several "green" fuels are therefore proposed in the industry to reduce emissions from ships. These include ammonia and hydrogen, which are considered potential maritime fuels for the future, but they both face various challenges related to toxicity, logistics, and infrastructure.

Green Methanol is another viable alternative fuel, offering several benefits over other options.

- Compatibility: There are already over 120 methanol-fueled vessels in operation (e.g. Maersk, CMA CGM)
- Handling Safety: Methanol is stored in ambient conditions and is less toxic than ammonia.
- Carbon Neutrality Potential: When produced from recycled CO_2 and green H_2.

Currently, the supply chain for green methanol is limited to less than 0.5 MTPA. However, the demand is expected to increase significantly, with some forecasts showing up to 30 Mt/yr demand by 2030.

The solution presented in this paper addresses this opportunity and comprises 3 elements.

a. Onboard Carbon Capture and Storage (OCCS) on ships, with CO_2 offloaded at port.
b. A low-cost green methanol production process, combining the CO_2 recovered from shipping with green hydrogen produced from renewable energy.
c. Packaging of this process plant, either as a modular facility for onshore sites or as an integrated FMPS facility, built in a shipyard and deployed to ports, or other sites, worldwide.

The basic scheme proposed is shown in the block diagram in Figure 1.

CO_2 (in orange) is recovered from the ship's main engine exhaust using an amine solvent wash system, after which the CO_2 is compressed, liquefied and stored in buffer storage on board.

In port, as part of the bunkering operation, the CO_2 is transferred to the feedstock storage of the green methanol plant. This may be supplemented by the import of CO_2 from other industrial sources.

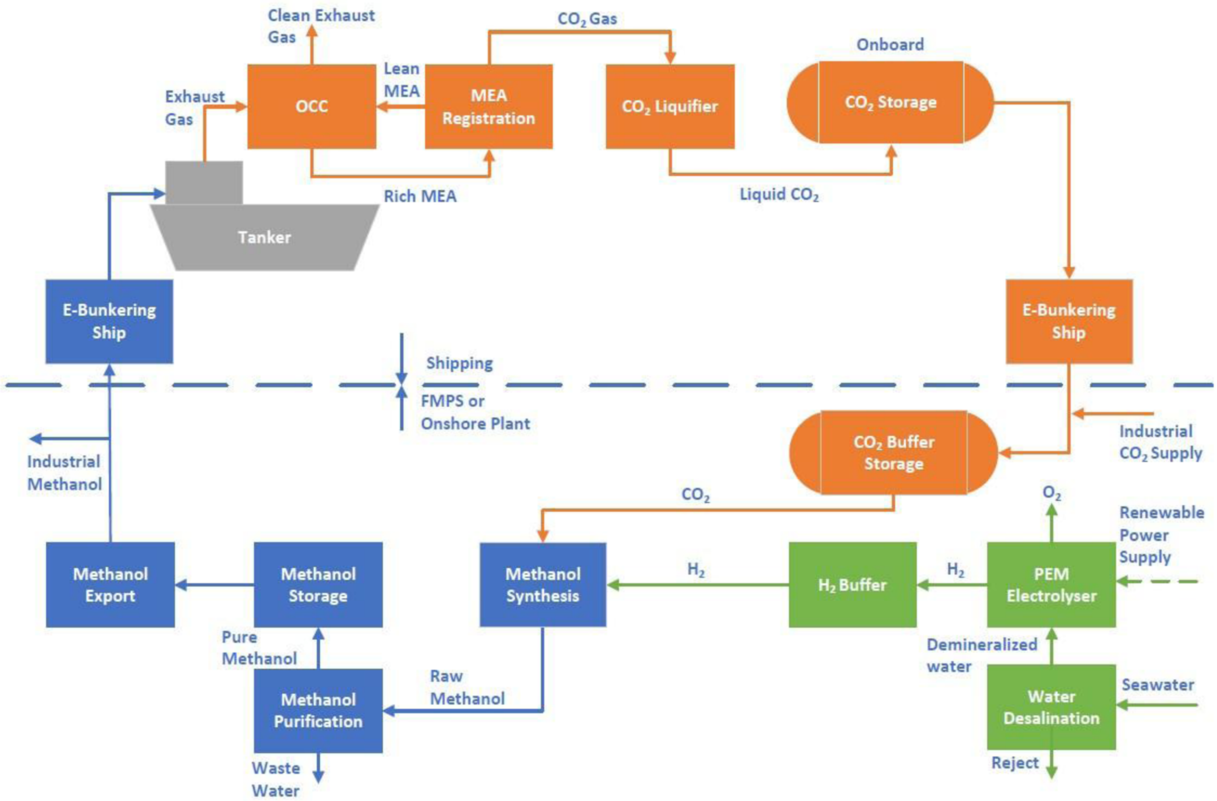

Figure 1—Block diagram for the Circular Green Methanol concept.

Green hydrogen is generated through the electrolysis of desalinated water powered by renewable energy sources such as solar, wind, or hydropower. The methanol synthesis process (in blue) takes CO_2 from storage, which is mixed with hydrogen and compressed to the pressure needed for the methanol synthesis reaction.

Raw methanol is purified through distillation and then stored for export to bunkering vessels or other industrial consumers.

Each step in this circular concept is discussed in more detail in the sections below.

Onboard Carbon Capture on Ships (OCCS)

Carbotreat BV develops and produces carbon capture systems for land-based, offshore, and maritime applications. This technology has been effectively installed in waste incineration plants for over a decade. The field-proven land-based solutions have allowed Carbotreat to "marinize" the system and apply it to seagoing vessels. In 2023, Carbotreat successfully demonstrated OCCS onboard an LNG carrier, followed by a trial in 2024 on an offshore heavy lift vessel, both as part of the EverLoNG program (EverLoNG, 2025)

Description of OCCS process

A schematic of the Carbotreat OCSS process is illustrated in Figure 2 and consists of 5 elements. First, the exhaust gas is cooled in a quench column. Then, the cooled exhaust gas is contacted with an amine solvent, such as MEA, which absorbs 80% to 90% of the CO_2. The cleaned exhaust gas exits via the top of the funnel into the atmosphere.

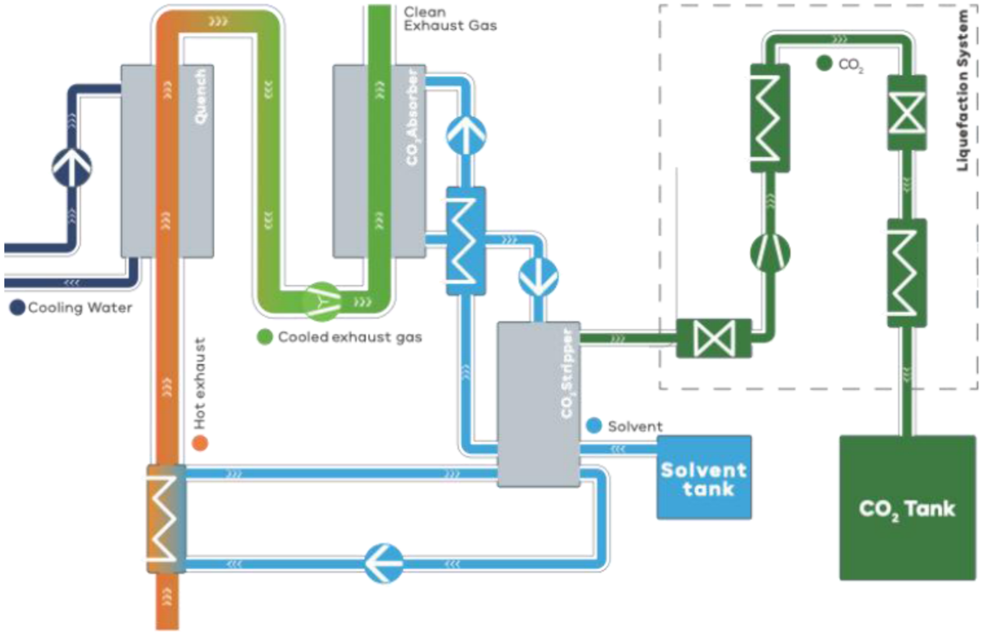

Figure 2—Schematic for Onboard Carbon Capture on Ships (OCCS).

In the 3rd step, the CO_2-rich solvent is regenerated by heat in the stripper column, making use of heat from the exhaust gas. This releases the CO_2 in gaseous form, after which the lean solvent is returned to the contactor in a closed-loop system.

Next, the gaseous CO_2 is liquefied by compression and cooling. This process produces over 99.9% pure liquid CO_2, which, in the final step, is stored in a tank onboard the ship. The high-purity CO_2 is then ready for permanent storage or utilization as feedstock for other products, such as synthetic fuels. The onboard CO_2 storage tank is sized depending on the required capture rate and the duration of the ship's voyages.

There are many different types of amine solvents available on the market, each with its own characteristics and benefits. The best solvent can be chosen for each application, and every project should therefore begin with a pre-engineering study to assess the flue gas specifications, the ship's engine data, and the sailing profile. Using this information, the most suitable amine solvent can be identified for each specific situation.

Status of OCCS Pilot Trials

Based on a decade of experience with land-based carbon capture systems, Carbotreat has adapted CCS technology for marine applications.

In the EverLoNG project, the OCCS system was successfully demonstrated onboard two vessels. Both campaigns were performed using the same benchmark post-combustion capture technology, chemical absorption of CO_2 by an amine solvent, which has been implemented onshore.

In the first trial campaign, the OCCS unit was installed onboard an LNG carrier (the Seapeak Arwa, which was chartered by TotalEnergies – Figure 3). In this application, the small-scale OCSS was installed adjacent to the funnel to demonstrate the capture technology. The trials achieved capture rates up to 80%.

Figure 3—LNG Tanker Seapeak Arwa showing the OCCS (in green) to the left of the funnel.

The second trial campaign was performed onboard the semi-submersible crane vessel Sleipnir (owned by Heerema), shown in Figure 4. Unlike the first trial, this one included both the capture and liquefaction of CO_2.

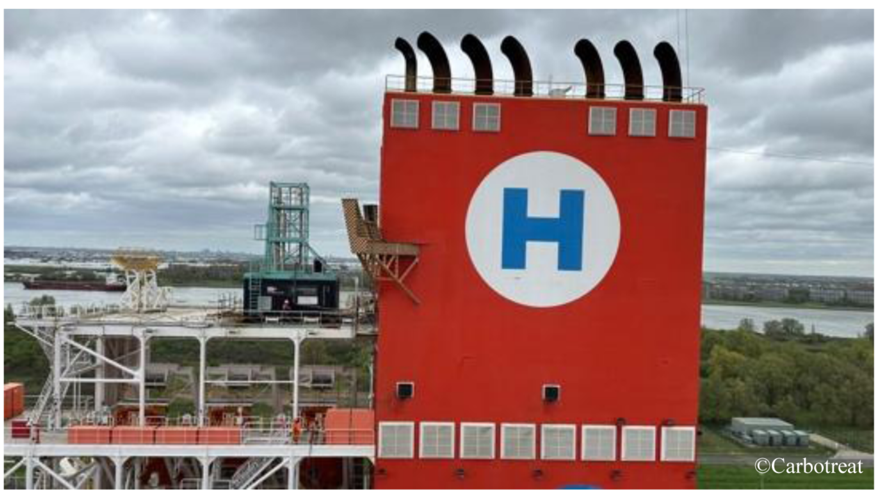

Figure 4—The Heerema Sleipner crane vessel, showing the OCCS (in green) to the left of the funnel.

These two campaigns enabled a detailed evaluation of the OCCS system's performance, particularly concerning the emissions of amines and ammonia (an amine degradation product) to the atmosphere, as well as the solvent degradation rate.

The amine emissions during the LNG carrier campaign were low and easily managed by the standard water wash system included in the CO_2 capture unit. However, amine emissions during the Sleipnir trial were somewhat higher, which was attributed to aerosols present in the exhaust gas.

Solvent degradation was relatively high during the LNG carrier campaign, due to the NO_2 content in the exhaust gas. However, for the Sleipnir campaign, NO_2 was removed from the exhaust gas before entering the amine absorber, which allowed degradation to be controlled to relatively low levels.

These learnings are essential to de-risk the future implementation of OCCS on ships.

Carbotreat's land-based capture systems are mature and have achieved TRL9 based on (*ISO 16290:2013*) and EU Horizon TRL definitions (Strazza, C., et al, 2017). With the transfer of this established technology to marine applications and successful outcomes from the two pilot trials, the OCCS technology has already reached TRL7 and is soon to advance to TRL8.

The latest project involves scaling up the technology to a full-scale OCCS unit. The project "Blue Horizon", part of the Dutch Maritime Masterplan, involves the manufacturing and installation of a 1.5 t/h OCCS system on an LNG carrier owned by Anthony Veder. The aim of this project is to reduce the vessel's CO_2 emissions by 70% and offset the remaining CO_2 emissions through the use of biofuels.

A key goal of this project is for the OCCS unit to develop a standard modular design, enabling the product to be utilized on other vessels as well. In future, retrofitting OCCS on older ships is expected to help them comply with new IMO Regulations, potentially allowing their service life to be extended in some cases.

Green Methanol Production (Onshore)

Whereas most methanol plants worldwide produce methanol from synthesis gas (CO and H_2), with typical capacities in the range 5,000 to 10,000 metric TPD, the process for direct hydrogenation of CO_2 by reaction with H_2 is much more recent and is gaining traction for the valorization of CO_2.

Wison Engineering Ltd (WEL), an EPC contractor based in Shanghai, China, has recently delivered a demonstration plant for methanol synthesis from CO_2 hydrogenation, namely the Waigaoqiao 3 project. A 3D view of this plant is shown in Figure 5 below. It has been constructed at the Waigaoqiao No. 3 coal-fired power station in China, with exhaust gas being the source of CO_2.

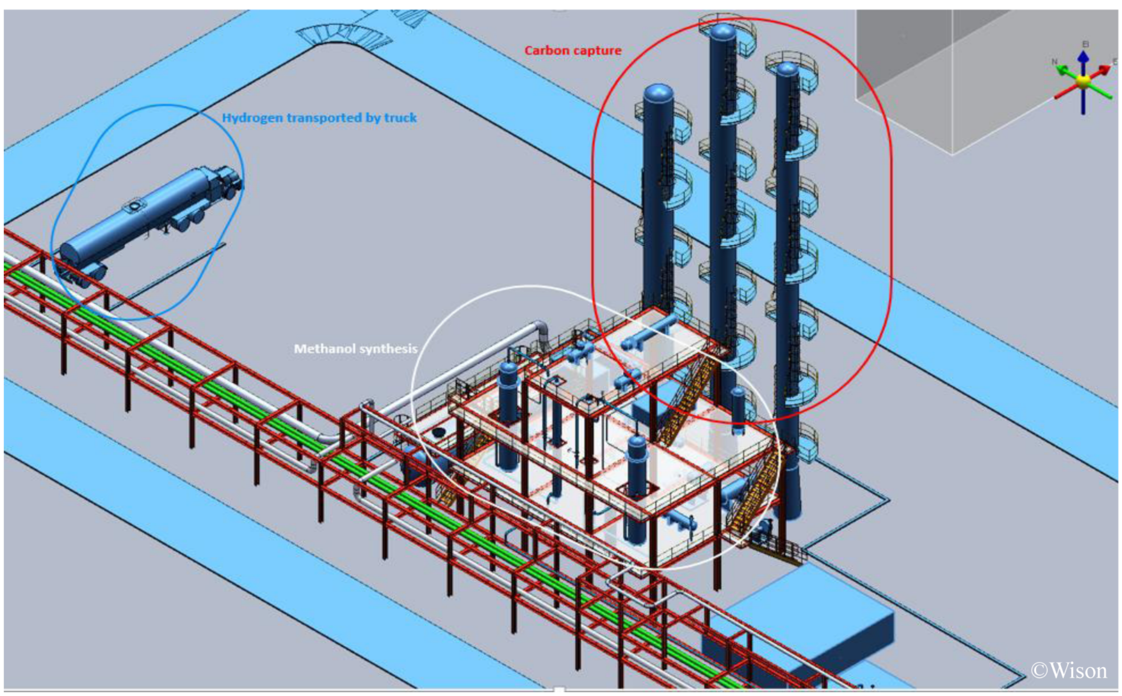

Figure 5—Schematic of methanol plant at Waigaoqiao No.3 coal-fired power station

The demonstration plant can produce 10,000 metric TPA of methanol and consists of a carbon capture unit and a methanol synthesis unit. CO_2 is captured from the flue gas of the power station through an amine scrubber system, and the H_2 is transported by truck to the site. CO_2 and H_2 are pressurized before being fed into a methanol synthesis reactor, which operates at around 220°C and 4.9 MPaG (49 Barg). The methanol is separated and exported, while the unreacted gas is recycled back to the reactor.

MegaFlex green Hydrogen Plants

WEL and Sungrow Hydrogen, a specialist in water electrolysis hydrogen production technology, have developed the "MegaFlex" concept for large-scale turnkey green hydrogen production. This transforms traditional green hydrogen plants into replicable, scalable, and rapidly deployable industrial products, significantly improving delivery efficiency and optimizing the levelized cost of green hydrogen (LCOH).

The MegaFlex plant (see Figures 6 and 7) combines the "Mega+Flex" design concepts - where "Mega" represents its 500MW single-unit hydrogen production capacity, and "Flex" demonstrates the solution's adaptability to scale up to GW-level projects or down to medium/small-scale applications, meeting diverse green hydrogen development needs across different phases. The integrated Wison-developed "RelPS" intelligent simulation platform optimizes configurations and control strategies based on renewable energy inputs, downstream load demands, and operational requirements, effectively reducing both initial investments and operating costs.

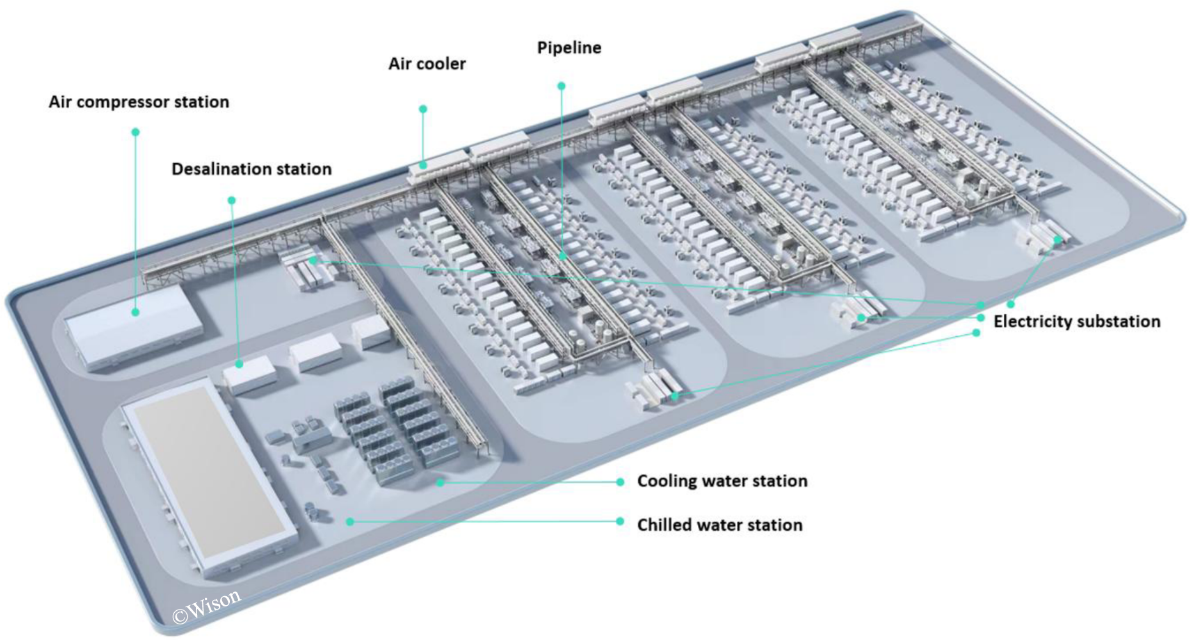

Figure 6—MegaFlex 3D view of Green Hydrogen Plant

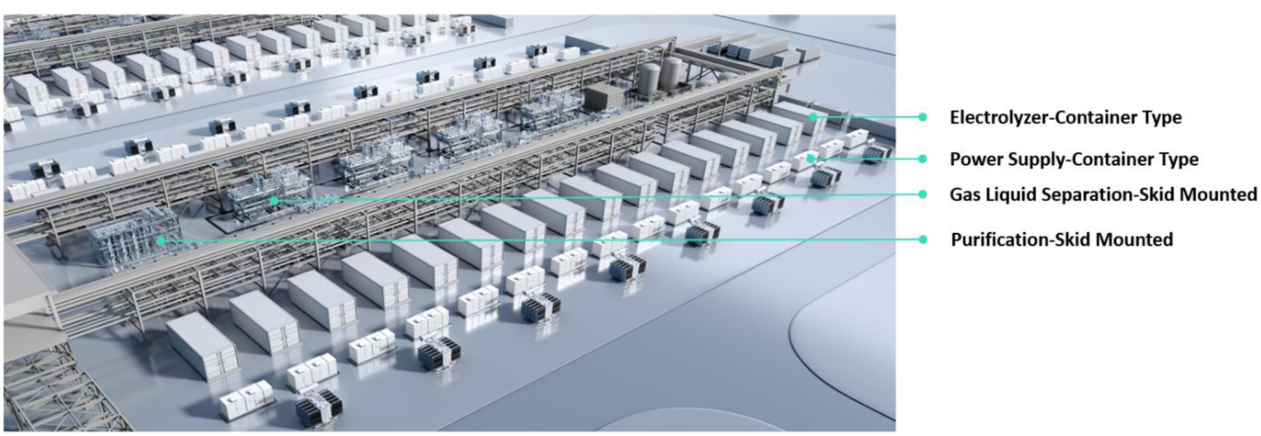

Figure 7—Electrolyzer Layout in MegaFlex Green Hydrogen Plant

The MegaFlex concept brings several key benefits for green hydrogen production.

- Proven Flexible Water Electrolysis Technology – Safe, reliable hydrogen production with smart digital management and global engineering expertise.
- Scalable & Fast Deployment – Standardized designs for all project sizes, enabling quick setup and easy expansion.
- Modular & Cost-Efficient – Reduces on-site risks, on-site construction costs, and accelerates delivery for faster ROI.

Adapting MegaFlex for Green Methanol

Green methanol is produced using renewable energy with no CO_2 emissions. It serves as a promising alternative to fossil fuels in various sectors, such as maritime transportation. Green methanol can include bio-methanol, which is synthesized from syngas generated by biomass gasification, and CO_2-methanol, which is produced by green hydrogen from an electrolyzer and the CO_2 captured from a circular combustion system (such as green methanol, biofuel or biomass).

The MegaFlex electrolysis system for green hydrogen generation has a short startup time and can quickly respond to renewable power's fluctuations or load changes. The hydrogen can be coupled with the captured CO_2 for CO_2-hydrogenation but also can be used as supplemental gas to adjust the hydrogen/carbon ratio of the syngas from biomass or biofuel gasification.

The modularity and scalability of MegaFlex allow this electrolysis system to support rapid expansion and quick deployment, meeting the hydrogen production needs of various scales and making it suitable for distributed hydrogen generation scenarios.

TRL assessment

MegaFlex system is based on the PEM electrolyzer, which is commercially developed and is proven at TRL 9, featuring a compact design, high power densities, short start-up time and quick response capabilities. As of July 2025, the world's largest PEM electrolyser project has gone into operation in Germany, where the electrolyser has an installed capacity of 54 megawatts and can produce 1 metric tonne of green hydrogen per hour (Fabian, 2025).

The technology for direct hydrogenation of CO_2 into methanol has been demonstrated and commercialized at TRL 9 by the large-scale green methanol plants around the world, whose capacities of methanol vary from 10,000 to 100,000 metric TPA.

Remaining challenges for scale-up

There are 3 main challenges to be overcome for the wider scale-up of green methanol to full commercial levels.

- High cost of green hydrogen production

  Hydrogen production costs are influenced by the CAPEX of the electrolyzer plant and the OPEX of the electricity cost, which accounts for around 80% of the cost of green hydrogen production. Lower electrolyzer prices and electricity prices are needed to make green hydrogen more competitive.
- Catalyst development

  In the process of methanol synthesis, the performance of the catalyst is crucial to the reaction efficiency and methanol yield. The catalysts need to be improved in activity, selectivity and stability. Considering that catalysts need to be replaced regularly, catalysts with lower costs and longer lifespans are also very important.
- Process integration and optimization with fluctuating renewable energy supply.

  The green methanol process involves multiple links, such as water electrolysis for hydrogen production, carbon dioxide capture, and methanol synthesis. Both solar and wind power are transient and can exhibit significant fluctuations in power output. This creates a fundamental challenge for maintaining a stable process system, since the methanol synthesis process demands a stable supply of H_2.

Ways to mitigate this can include a Battery Energy Storage System (BESS) upstream of the electrolyzer, selecting an electrolyzer capable of accepting fluctuating load changes, buffer storage of hydrogen upstream of the synthesis loop, and an intelligent planning & energy management system for capacity planning and energy dispatch optimization. Most likely, a combination of all these will be required.

Adaptation to a Green Methanol Floater (FMPS)

Wison New Energies (WNE) is an EPCIC supplier of floating energy systems and modularized onshore process plants, with a long track record in complex floaters such as FLNG vessels.

In recent years, WNE has developed concepts for floating green ammonia, floating blue methanol and floating green methanol. An illustration of the floating green methanol production unit is shown in Figure 8 below.

Figure 8—Floating Green Methanol Production, Storage and Offloading Unit (FMPS)

This concept has a capacity of approximately 300,000 metric TPA of green methanol, which requires around 300MW of electrolyzer capacity installed on the topside. CO_2 is delivered directly from bunkering ships, which receive it from intermediate storage on trading ships, or an onshore buffer storage system.

The main features of this FMPS example are:

a. Topside processing comprises 2 banks of electrolyzer modules, methanol synthesis and purification modules, and associated utilities.
b. The hull includes 100,000 m^3 of methanol storage, plus buffer storage for CO_2 and hydrogen. The hull in this case has dimensions of around 315m × 60m, with a depth of 28m.

c. The marine systems include crew quarters, a machinery space for utility systems and hotel services, and a methanol offloading system.

This integration of the green methanol process draws heavily on experience and best practices from the global FPSO and FLNG industries.

This concept has been developed to TRL 5 level (based on ISO16290) and is ready to commence a project-specific FEED study to reach TRL6. The main challenges to be overcome in FEED are as follows.

a. Layout. The multiple electrolyzers dominate the layout and must be stacked to enable sufficient capacity to be installed. Moving to a smaller number of larger electrolyzers, as these become available on the market, will be advantageous.
b. Motion. Although the FMPS will likely be moored at quayside, where motions will be low, the equipment must be validated for performance under low levels of dynamic roll and pitch, as well as static trim and list under varying tank loading conditions. This includes the electrolyzer plant, the methanol synthesis reactor, and the large methanol distillation column, none of which have yet been deployed on a floater.
c. Hull. Optimization of the new build hull design to reduce cost. The alternative of converting a suitable tanker can also be considered.

The FMPS has many potential benefits compared to a conventional onshore plant, which include:

- Lower costs in many applications, where an FMPS built in a high-efficiency shipyard will be cheaper than a stick-built onshore plant, especially in remote areas or locations with high labor costs.
- Schedule, since an FMPS can be fully built and commissioned in a shipyard at least 1 year faster than the equivalent onshore plant.
- Quality of the construction and commissioning work carried out in a controlled shipyard environment, using a skilled and trained workforce, can be better than a transient workforce hired for an onshore plant.
- Safety, which can benefit from a controlled shipyard environment in the same way as Quality, above.
- The FMPS concept offers the potential to standardize and replicate the product, thereby reducing costs through the "Design One, Build Many" approach, which has been successful in some FPSO and FLNG applications.
- The FMPS can be relocated to another site if market conditions change.
- The FMPS could adopt similar commercial models used in the FPSO and FLNG markets, such as the lease & operate approach, instead of outright purchasing on an EPC or EPCIC basis.
- The FMPS does not require onshore land space, which can be at a premium in a port location.

The first 5 bullet points are also partly applicable to an onshore plant which is modularized and shipped to a remote onshore site for erection and integration. The selection of the optimum deployment strategy will therefore be project-specific, depending on the project location and local factors.

Deployment Scenarios

We envisage 3 main deployment scenarios for this circular green methanol concept. All assume that the facility would be situated close to a shipping transport hub, to facilitate the bunkering of methanol fuel onto the ships, and at the same time collecting the captured CO_2 from the OCCS storage on the ships. These 3 deployment scenarios are:

1. Land-Based Hub
2. Floating Hub
3. Hybrid Land + Floating Hub

The land-based facility could be used where there is suitable land available for the green methanol plant and where skilled construction labor is readily accessible at reasonable cost. In this case, the main process plant may be designed and constructed as a modularized plant, with the modules being built and pre-commissioned in a controlled shipyard environment.

The floating FMPS facility may be preferred where there is a lack of land in the vicinity of the shipping transport hub, or where construction labor is in short supply, or prohibitively expensive. In this case, the entire green methanol plant can be designed and built as a floating facility, which is built in a shipyard and towed (or sailed) to the final destination. This will typically result in the lowest CAPEX and project schedule, but at the expense of local content in the host country.

A hybrid combination of these two deployment scenarios can also be interesting, where a smaller floating facility is deployed as an Early Production System (EPS) to cover the initial phase of the growing market demand and also to provide early cash flow to the project. As the demand grows, a larger onshore plant can be built in parallel with the EPS operation. Once the onshore plant is fully operational, the EPS may be relocated to a second site.

A further option could be to split the green hydrogen production, which requires significant space for the electrolyzer plant, from the methanol plant. The green hydrogen could be produced onshore, and the FMPS restricted to methanol synthesis, purification, and storage. This could be interesting for several scenarios, such as:

- Where there is already a source of green (or blue) hydrogen available from an existing plant.
- Where there is land space available onshore, the electrolyzer plant could be located onshore allowing a larger capacity of methanol synthesis and purification to be installed on the FMPS. (The electrolyzer plant could still be delivered as complete modules.)

Economics

The economics of the circular green methanol concept depend on 3 main elements.

a. CAPEX and OPEX of the plant
b. Cost of green power (electricity)
c. Cost (or transfer cost) of CO_2 feedstock

To illustrate this, we show below a sensitivity analysis for a typical deployment scenario. This assumes the following design basis for the project.

- Capacity 300,000 metric TPA of green methanol
- Power demand 300 MW
- Complete, highly-standardized FMPS facility built in a low-cost shipyard

The CO_2 transfer cost must justify the installation of OCCS plants, considering a typical 80% recovery rate of CO_2 from the exhaust stack. Factors such as vessel life extension, carbon taxes, and regional subsidies may impact the transfer pricing calculation.

The green power electricity cost will vary significantly between regions, depending on factors such as environmental conditions and local subsidies available.

For these reasons, we have shown the costs of CO_2 and power as a broad range and calculated the breakeven price of green methanol for each scenario. Figure 9 below shows that the range of methanol prices needed to break even falls between \$350/T and \$800/T, depending on the assumptions for input costs.

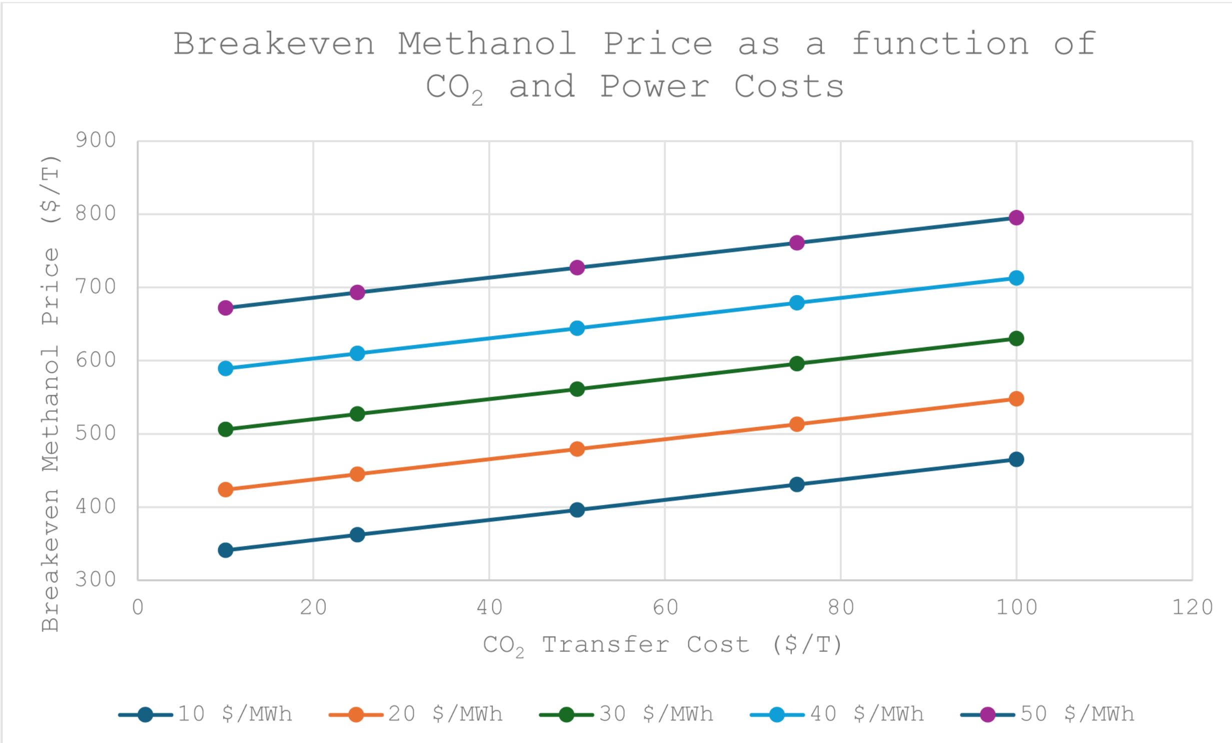

Figure 9—The breakeven Methanol price as a function of CO_2 transfer cost and the electricity power cost.

In context, the current price of methanol in Europe and the USA ranges from \$500/T to \$750/T and is forecast to surpass \$1000/T by 2030.

We therefore believe that there can be an attractive business opportunity for the circular green methanol concept, especially in locations where low-cost renewable power is available. The key to making this solution economic is to adopt a highly standardized floating FMPS facility design, which can follow the "Design One, Build Many" approach using a low-cost, high-quality shipyard.

Conclusions

The work described in this paper confirms that the integration of Onboard Carbon Capture on Ships (OCCS), coupled with green methanol production on an FMPS vessel, establishes a maritime fuel cycle that is technically feasible, economically viable and scalable.

No technology roadblocks were identified, and the concept is ready to proceed to a FEED level study. The OCCS system for capturing CO_2 on ships has been proven by Carbotreat at pilot demonstrator scale and is now being implemented on the first commercial project. The green methanol process, using direct hydrogenation of CO_2 captured from exhaust gas, is also proven at pilot demonstrator scale by Wison Engineering and is ready for scale-up to full commercial application. Marinizing this process for installation on a floater has been studied by Wison New Energies and has been found to be feasible, although it requires further work to optimize the integration and solve some of the detailed engineering challenges, mainly related to the impact of motions. Lessons learned from FPSO and FLNG projects are invaluable for that purpose.

The economics of the FMPS are attractive for sites where the cost of electricity and the transfer cost of CO_2 are low. If the market price of green methanol increases towards $1000/T, as forecast by some, then the economics could become compelling. There are significant opportunities for standardization of replication of the FMPS (Design One, Build Many), which would help to drive down the CAPEX and further improve the project economics.

There are opportunities for deployment of standardized FMPS units across many global shipping hubs to supply a significant share of the future global bunker demand. Once standards are fixed for CO_2 feedstock quality and the renewable power supply voltage and frequency, it would be feasible to have a truly standardized product to suit a wide range of global sites.

The FMPS is the first floating methanol product under development by Wison. As the market expands, we anticipate many opportunities for further innovation in this field.

Glossary

BESS	Battery Energy Storage System
CAPEX	Capital Expenditure
EPC	Engineer, Procure, Construct form of contract
EPCIC	Engineer, Procure, Construct, Install and Commission form of contract
FMPS	Floating Green Methanol Production, Storage and Offloading vessel
FPSO	Floating Production, Storage and Offloading vessel
FLNG	Floating LNG Liquefaction, Storage and Offloading vessel
GHG	Greenhouse Gas
IMO	International Maritime Organisation
MEA	Monoethanolamine
TPA	Tonnes per Annum
TPD	Tonnes per Day
MTPA	Million Metric Tonnes per Annum
OCCS	Onboard Carbon Capture on Ships
OPEX	Operating Expenditure
PEM	Proton Exchange Membrane electrolyzer
ROI	Return on Investment
WEL	Wison Engineering Ltd
WNE	Wison New Energies

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